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*Modeling light propagation in photonic nanostructures
using the finite element method*

Bachelor's thesis
in nanostructure engineering
in the field of photonics

Thesis prepared under the supervision of
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Statement of the thesis supervisor

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Abstract

The finite element method was applied to model the propagation of light in periodic photonic nanostructures. For this purpose, a two-dimensional vector wave equation was solved. Simulations were performed for photonic crystals with different lattice structures, into which linear defects were introduced. Differences resulting from the use of waves of different frequencies, lying both inside and outside the photonic band gap of the system, are presented. A simulation of guiding a localized mode in the linear defect of a photonic crystal was performed. The transmission of electromagnetic radiation through diffraction gratings with double periodicity was also studied. Modeling was carried out for systems made of silver, gold, chromium, and a perfect conductor, for a wavelength of 400 nm.

Keywords

Finite element method, electromagnetic modeling, PML, light, vector wave equation, photonic crystal, photonic band gap, subwavelength grating

Field of the thesis (codes according to the Socrates-Erasmus program)

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Title of the thesis in English

Modeling of light propagation in photonic nanostructures using the finite element method

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Chapter 1. Introduction

Photonics is a rapidly developing field of science that combines elements of optics, solid-state physics, materials engineering, computer science, and electronics. The ability to control light has become extremely important today, among other things for efficient information transfer, more accurate imaging, and the production of advanced optical filters. The newest discoveries in condensed matter physics became a starting point for research on the use of semiconductors in optics, which brought new possibilities that had not been known before. Dielectric micro- and nanostructures, such as photonic crystals, used by nature for millions of years, have made possible, among other things, selective guiding and trapping of light, while metal-dielectric systems have become the basis for the development of plasmonics.

Because of the cost and the difficult-to-predict properties of complex photonic systems, it is necessary – in order to obtain the desired properties and optimize them – to carry out computer simulations. Well-developed methods for studying light propagation in new materials include, for example, FEM (finite element method) and FDTD (finite difference time domain method). To determine the band structure of periodic systems, the PWM method (plane wave method) can be used. The combination of these methods makes it possible to obtain a wide range of information about the structure under consideration, which enables possible modifications to improve efficiency and also to find new potential applications.

This thesis is based on the finite element method, one of the methods for solving differential equations and an extremely useful tool for modeling the behavior of electromagnetic waves in photonic structures. It was used to study light propagation in photonic crystals with different lattice types and different sizes. It was also applied to determine the properties of diffractive elements with double periodicity, depending on the direction from which the structure is illuminated and on the material from which it is made.

In summary, the use of computer modeling to design new nano-optical systems is an important problem that enables the production of modern photonic devices. The foundations of finite element method simulations, supplemented by the finite difference time domain method and the plane wave method, are presented in this thesis.

Chapter 2. Theoretical background

2.1. Maxwell equations and the two-dimensional vector wave equation

Electromagnetic fields in vacuum are uniquely described by the four Maxwell equations, whose differential form is:

$$\nabla \cdot E = \frac{\rho}{\epsilon_0}, \quad (2.1a)$$

$$\nabla \times E = -\frac{\partial B}{\partial t}, \quad (2.1b)$$

$$\nabla \cdot B = 0, \quad (2.1c)$$

$$\nabla \times B = \mu_0 J + \mu_0 \epsilon_0 \frac{\partial E}{\partial t}. \quad (2.1d)$$

The vector $E \equiv E(r, t)$ denotes the electric field intensity, the vector $B \equiv B(r, t)$ is called the magnetic induction, and J is the electric current density. The quantities ρ , ϵ_0 , and μ_0 are, respectively, the electric charge density, the magnetic permeability of vacuum, and the electric permittivity of vacuum.

Maxwell equations take an analogous form in material media and are presented below.

$$\nabla \cdot D = \rho_{\text{sw}}, \quad (2.2a)$$

$$\nabla \times E = -\frac{\partial B}{\partial t}, \quad (2.2b)$$

$$\nabla \cdot B = 0, \quad (2.2c)$$

$$\nabla \times H = J_{\text{sw}} + \frac{\partial D}{\partial t}. \quad (2.2d)$$

The vector D is the electric displacement, H is the magnetic field intensity, J_{sw} is the density of free currents, and ρ_{sw} is the density of free electric charges.

The fields D and H are related to the fields E and B by constitutive equations that depend on the properties of the given substance. For optically linear materials:

$$D = \epsilon \epsilon_0 E, \quad (2.3)$$

$$H = \frac{1}{\mu \mu_0} B, \quad (2.4)$$

where ϵ and μ are dimensionless relative electric permittivity and magnetic permeability, dependent on the material in which Maxwell equations are solved. In addition,

$$J_{\text{sw}} = \sigma E \quad (2.5)$$

is known as Ohm's law, where σ is the specific electrical conductivity [1].

Assuming the absence of sources, linearity of the medium, monochromaticity of the sought electromagnetic waves, and introducing complex notation ($E(r, t) = \Re(\tilde{E}(r) \cdot e^{i\omega t})$, $H(r, t) = \Re(\tilde{H}(r) \cdot e^{i\omega t})$, where $\Re$ denotes the real part and ω is the frequency of the wave), one obtains Maxwell equations in the frequency domain.

$$\nabla \cdot (\epsilon \tilde{E}) = 0, \quad (2.6a)$$

$$\nabla \times \tilde{E} = -i\omega \tilde{B}, \quad (2.6b)$$

$$\nabla \cdot \tilde{B} = 0, \quad (2.6c)$$

$$\nabla \times \tilde{H} = i\omega \epsilon \epsilon_0 \tilde{E}. \quad (2.6d)$$

From the above identities one can next isolate a relation known as the vector wave equation, where $k_0 = \omega/c$ is the wave number and $c = 1/\sqrt{\epsilon_0\mu_0}$ is the speed of light in vacuum.

$$\nabla \times \epsilon^{-1} \nabla \times \tilde{H} = k_0^2 \mu \tilde{H}. \quad (2.7a)$$

The vector wave equation can also be written in a form containing the electric field intensity vector:

$$\nabla \times \mu^{-1} \nabla \times \tilde{E} = k_0^2 \epsilon \tilde{E}. \quad (2.7b)$$

If the medium under consideration is anisotropic, ϵ and μ take the form of diagonal matrices (tensors) with coefficients specifying the electric permittivities and magnetic permeabilities for all distinguished directions. Restricting the discussion to the diagonal form of anisotropy gives

$$\epsilon = \begin{bmatrix} \epsilon_x & 0 & 0 \\ 0 & \epsilon_y & 0 \\ 0 & 0 & \epsilon_z \end{bmatrix}, \quad (2.8a)$$

$$\mu = \begin{bmatrix} \mu_x & 0 & 0 \\ 0 & \mu_y & 0 \\ 0 & 0 & \mu_z \end{bmatrix}. \quad (2.8b)$$

Finally, for two-dimensional problems in which the partial derivatives of the electric field intensity, magnetic field intensity, electric permittivity, and magnetic permeability vanish in the z direction, and which were considered during the simulations presented later in this thesis, two equations are obtained. For TE (Transverse Electric) and TM (Transverse Magnetic) polarization, respectively, one has:

$$\nabla \cdot (\mu^{-1} \nabla E_z) + k_0^2 \epsilon_z E_z = 0, \quad (2.9)$$

$$\nabla \cdot (\epsilon^{-1} \nabla H_z) + k_0^2 \mu_z H_z = 0. \quad (2.10)$$

These are elliptic differential equations. In the first case the oscillations of the electric field are perpendicular to the plane in which the simulation is performed. In the second case, the oscillations of the magnetic field are perpendicular to it. It is worth mentioning that cases are found in the literature in which the naming of these polarizations is reversed.

2.2. Finite element method

The finite element method (FEM) is a method widely used in physics for solving differential equations, including for systems in which several physical processes occur simultaneously.

The method is based on discretizing the region in which the solution is sought into simple elements, which in general have an irregular shape. In two-dimensional problems, a triangular mesh is most commonly encountered, as shown in Fig. 2.1.

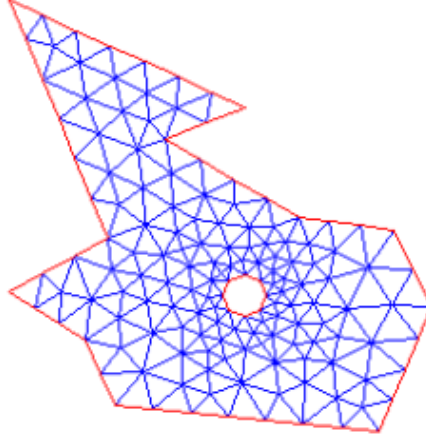


Fig. 2.1. Example simulation region divided by an irregular triangular mesh.

The solution of the differential equation is the sum of approximate solutions found for all mesh elements [2].

$$f(x, y) = \sum_{i=1}^n f_i(x, y). \quad (2.11)$$

In the simplest case, the functions $f_i(x, y)$ have the form of first-degree polynomials of two spatial variables:

$$f_i(x, y) = a + bx + cy. \quad (2.12)$$

Finding the coefficients a , b , and c requires solving a system of three linear equations with three unknowns, whose coefficients are combinations of the positions (x_{iw}, y_{iw}) of the nodes of the element and of the solutions of the given differential equation at those points. The matrix form of the system of equations is presented below.

$$\begin{bmatrix} f_{1w} \\ f_{2w} \\ f_{3w} \end{bmatrix} = \begin{bmatrix} 1 & x_{1w} & y_{1w} \\ 1 & x_{2w} & y_{2w} \\ 1 & x_{3w} & y_{3w} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}. \quad (2.13)$$

Thus the matrix of coefficients a , b , and c is given by

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 & x_{1w} & y_{1w} \\ 1 & x_{2w} & y_{2w} \\ 1 & x_{3w} & y_{3w} \end{bmatrix}^{-1} \begin{bmatrix} f_{1w} \\ f_{2w} \\ f_{3w} \end{bmatrix}. \quad (2.14)$$

Numerical methods for finding the values f_1 , f_2 , and f_3 depend strictly on the type of differential equation being solved.

2.3. Elliptic differential equation and a method for its approximate solution

The form of an elliptic differential equation is described by equation (2.15), where f is the function sought in the region Ω , and a , b , and c are coefficients of the equation.

$$-\nabla \cdot (c \nabla f) + af = b. \quad (2.15)$$

The value of f on the boundary of the region Ω (hereafter $\partial\Omega$) is specified by Dirichlet conditions of type (2.16a) or Neumann conditions of type (2.16b).

$$hf = r, \quad (2.16a)$$

$$\mathbf{n}(c \nabla f) + qf = g. \quad (2.16b)$$

The quantities h , r , and q are successive coefficients of the equations, and $\mathbf{n}$ is the normal vector to $\partial\Omega$. In the case of the two-dimensional vector wave equation, boundary conditions are chosen so as to avoid, as well as possible, reflections at the boundary of the simulation region. This problem can largely be avoided by using a PML (perfectly matched layer), which is discussed in section 2.4.

Equation (2.15) is multiplied by a test function v and both sides are integrated over the region Ω , yielding equation (2.17), and is then integrated by parts, bringing it to the form shown in (2.18) [3,4].

$$\int_{\Omega} ((-\nabla \cdot c \nabla f)v + au f) dx = \int_{\Omega} bv dx, \quad (2.17)$$

$$\int_{\Omega} ((c \nabla f) \cdot \nabla v + afv) dx - \int_{\partial\Omega} (-qf + g)v ds = \int_{\Omega} bv dx. \quad (2.18)$$

Next, the above equation is written in the so-called weak form, and its solution is called a weak solution. One seeks functions f satisfying (2.19) for all v .

$$\int_{\Omega} ((c \nabla f) \cdot \nabla v + afv - bv) dx - \int_{\partial\Omega} (-qf + g)v ds = 0. \quad (2.19)$$

The functions f and v belong to the same space V . One must choose a finite-dimensional subspace $V' \subset V$, whose dimension $\dim V'$ determines the number of nodes, and to which the set of test functions v_i forming the basis will belong. The dimension of V' determines the accuracy of the calculation result. The sought solution of the problem is the element of the subspace lying closest to the weak solution in the sense of the norm $\|r\|$, whose form is given below.

$$\|r\| = \int_{\Omega} (c|f|^2 + af^2) dx + \int_{\partial\Omega} qf^2 ds. \quad (2.20)$$

The function $f(x)$ can then be expanded in a series, giving the approximate solution

$$f(x) \approx \sum_{j=1}^{\dim V'} F_j v_j(x). \quad (2.21)$$

Substituting the above formula into the discretized equation (2.18) gives the following system of equations:

$$\begin{aligned} \sum_{j=1}^{\dim V'} \left(\int_{\Omega} ((c \nabla v_j) \cdot \nabla v_i + av_j v_i) dx + \int_{\partial\Omega} qv_j v_i ds \right) F_j \\ = \int_{\Omega} bv_i dx + \int_{\partial\Omega} gv_i ds, \quad i = 1, \dots, \dim V'. \end{aligned} \quad (2.22)$$

The quantity (2.23a) is called the stiffness matrix, while the quantity (2.23b) is called the mass matrix. These names follow from the applicability of the above method to solving problems in statics.

$$K_{i,j} = \int_{\Omega} (c \nabla v_j) \cdot \nabla v_i dx, \quad (2.23a)$$

$$M_{i,j} = \int_{\Omega} av_j v_i dx. \quad (2.23b)$$

The resulting system of equations of the form $AF = B$ gives the solutions of the elliptic differential equation at all nodes of the mesh. More extensive information on approximate methods for finding solutions of differential equations is presented in [3,4].

2.4. PML boundary conditions

A serious problem encountered when performing electromagnetic simulations is the reflection of propagating waves from the boundaries of the simulation region, resulting from imperfections of the boundary conditions used. This can seriously disturb the obtained results. To avoid this problem, a nonreflecting and lossy layer, called a PML (perfectly matched layer), is introduced [5].

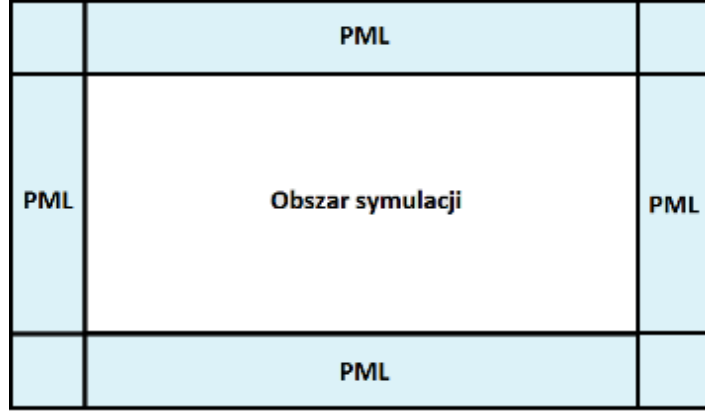


Fig. 2.2. Schematic of a system surrounded by a PML layer.

For a normally incident wave, the condition for the absence of reflections at the interface of media is equality of impedances, $\eta = \sqrt{\mu/\epsilon}$.

$$\sqrt{\frac{\mu_0\mu_1}{\epsilon_0\epsilon_1}} = \sqrt{\frac{\mu_0\mu_2}{\epsilon_0\epsilon_2}}. \quad (2.24)$$

A generalization of this condition to oblique incidence is an artificially introduced region with specially chosen anisotropic permittivities ϵ and permeabilities μ . A PML does not belong to the region of the considered problem and is nonphysical. Materials with properties close to those of a perfectly matched layer can, however, be obtained in the form of multilayer structures [6]. Finally, introducing the matrix

$$S = \begin{bmatrix} s^{-1} & 0 \\ 0 & s \end{bmatrix}, \quad (2.25)$$

$s = 1 + \alpha i$, where α is an arbitrarily chosen real number, the two-dimensional vector wave equations for nonmagnetic materials in PML regions for TE and TM polarization at the horizontal boundary of the simulation region take the forms (2.26a) and (2.26b), respectively, while in regions at the vertical boundary they take the forms (2.27a) and (2.27b).

$$-\nabla \cdot (S \nabla E_z) - s k_0^2 \epsilon_z E_z = 0, \quad (2.26a)$$

$$-\nabla \cdot (S \epsilon^{-1} \nabla H_z) - s k_0^2 H_z = 0, \quad (2.26b)$$

$$-\nabla \cdot (S^{-1} \nabla E_z) - s k_0^2 \epsilon_z E_z = 0, \quad (2.27a)$$

$$-\nabla \cdot (S^{-1} \epsilon^{-1} \nabla H_z) - s k_0^2 H_z = 0, \quad (2.27b)$$

where the operator ∇ is restricted to two dimensions. These equations can also be modified for corner regions; however, using one of the above also gives satisfactory results.

An example of the use of a perfectly matched layer is shown in Fig. 2.3. Diffraction of a TE-polarized electromagnetic wave by a narrow slit cut in a material that is a perfect

conductor was considered. Thanks to the use of PML boundary conditions, the absence of reflections can be observed. The result below was obtained using the FEM method.

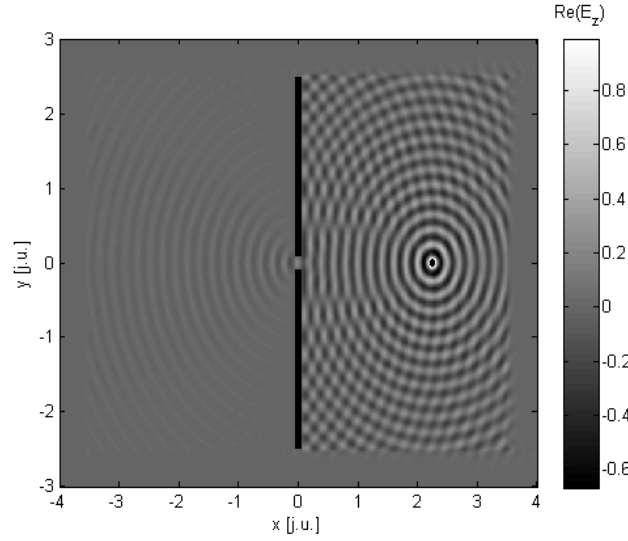


Fig. 2.3. Diffraction of an electromagnetic wave by a narrow slit.

2.5. Photonic crystals

A crystal is a periodic structure composed of atoms or molecules. The arrangement of these components defines the type of crystal lattice of the system. A characteristic property of crystals is the possible occurrence of a so-called band gap. This means that, for certain values of energy and momentum, an electron cannot move in such a structure. When propagation is impossible in any direction within a specified energy range, the gap is called complete; otherwise it is called partial. An example of a complete band gap is the band structure of semiconductors, in which it separates the valence band from the conduction band.

The analogous structure in optics is a photonic crystal. It is a periodic structure made of dielectric materials that differ in electric permittivity. The period of such a system is comparable with the wavelength for which it is to perform its function. Depending on the directions of periodicity, one-, two-, and three-dimensional crystals are distinguished.

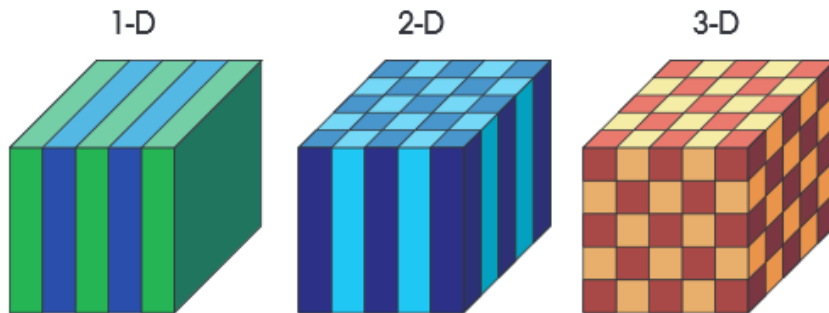


Fig. 2.4. Schematic of one-, two-, and three-dimensional photonic crystals. Differences in color symbolize materials with different values of electric permittivity [7].

As in ordinary crystals (or generally, in periodic media), the modes of photonic crystals have the form of Bloch waves. It also happens that electromagnetic radiation of certain frequencies cannot propagate in the structure. This is called a photonic band gap. If, for a given frequency range, the gap occurs for all polarizations and propagation directions, it is called complete; otherwise it is partial.

The possibility of selective propagation of light makes it possible to create structures with interesting optical properties, such as mirrors reflecting electromagnetic radiation in a specified wavelength range, materials with nearly zero group velocity ("slowing" light), opposite values of group and phase velocity (backward propagation), or materials at whose boundary negative refraction occurs. Another possibility is the creation of waveguides based on photonic crystals. A linear defect is introduced into the structure by removing a row or several rows of dielectric rods placed in a homogeneous matrix. In the range of the photonic band gap, a defect mode appears, allowing the propagation of electromagnetic radiation along the resulting waveguide. An advantage of such a structure is its much lower loss, for example in the visible-light range, compared with metallic waveguides.

2.6. Diffraction gratings

A diffraction grating is a periodic system of slits (apertures) or obstacles that causes changes in the intensity, phase, or both of a transmitted or reflected wave. Two basic types of gratings are distinguished: amplitude gratings and phase gratings. In the first case, light encounters alternating transparent and opaque regions, which modulates the amplitude. In the second case, the optical path difference resulting from corrugation of transparent elements modulates the phase. If the corrugated diffraction grating is not completely transparent in places, both amplitude and phase modulation occur.

For a wave normally incident on a transmission grating, one can introduce the equation

$$\Lambda \sin \theta = m\lambda, \quad (2.28)$$

relating the grating period Λ , the angle θ at which intensity maxima are observed, and the wavelength λ . The integer m indexes the diffraction order. The above equation can be generalized for oblique incidence [8]:

$$\Lambda(\sin \theta_m - \sin \theta_i) = m\lambda. \quad (2.29)$$

θ_i is the angle at which the wave is incident on the structure, and θ_m is the angle at which the m -th diffraction order is observed.

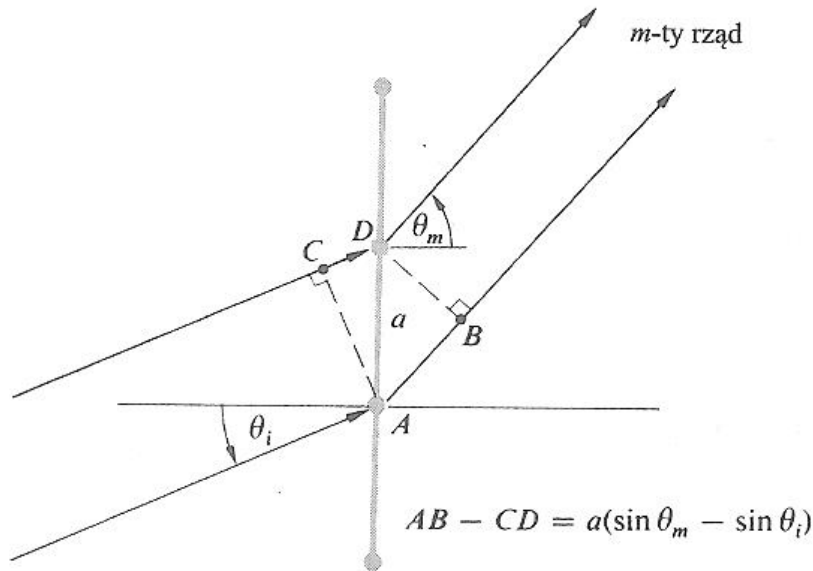


Fig. 2.5. Schematic of oblique incidence of electromagnetic radiation on a transmission diffraction grating [8].

The above relations show that, for a given wavelength and a specified grating period, there is a finite number of real diffraction orders. It is known that, in two dimensions,

$$k_x^2 + k_y^2 = k_0^2 n^2, \quad (2.30)$$

where n is the refractive index of the external medium. At the same time, for a wave diffracted by a system periodic in the x direction, the vector k_x is conserved up to multiples of $2\pi/\Lambda$:

$$k_x^{(0)} = k_x^{(i)} + \frac{2\pi m}{\Lambda}, \quad (2.31)$$

where the index (i) corresponds to the incident wave and (0) to the wave leaving the grating. Thus, for large m , $|k_x^{(0)}| > k_0 n$, and therefore $k_y^{(0)} = \sqrt{k_0^2 n^2 - (k_x^{(0)})^2}$ takes imaginary values. One then deals with evanescent, exponentially decaying diffraction orders.

In the case of diffraction gratings composed of several systems of elements with different periods, the number of real diffraction orders depends on the direction from which such a structure is illuminated. This is shown in the simulations whose results are presented in section 3.2.

Chapter 3. Simulation results

This chapter presents simulations carried out independently. The finite element method was used (the "PDE Toolbox" package by MathWorks), and its results were compared with selected results obtained using the finite difference time domain method [9] (the MEEP program [10]). To find the band structure of photonic crystals, the MPB – MIT Photonic Bands program [11], based on the plane wave method, was used.

3.1. Propagation of light in photonic crystals

Two-dimensional photonic crystals with two lattice types were considered: square and triangular.

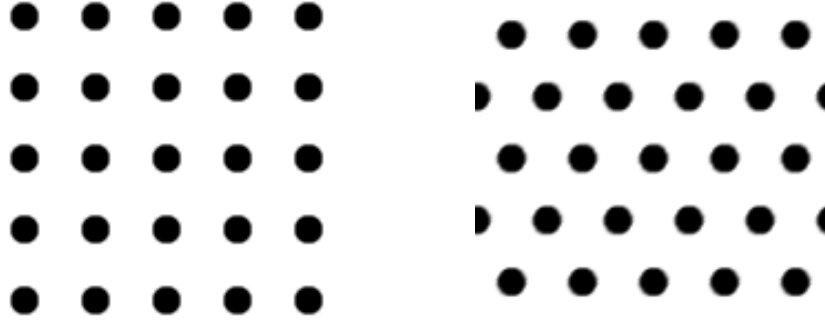


Fig. 3.1. Schematics of two-dimensional crystal lattices: square (left) and triangular (right).

Each black element in the above figure is a dielectric rod of radius $0.2a$ (where a is the lattice period) and $n_1 = 1.7865$, which is the refractive index of sapphire (Al_2O_3) for electromagnetic radiation of wavelength 400 nm [12]. The rods were placed in a material with $n_2 = 1$, which with good accuracy can be taken as the refractive index of air (in fact, it is the refractive index of vacuum).

Band structures were determined for a specified, previously defined refractive index. However, because of dispersion, this index actually changes with wavelength [13]. Therefore, modeling was performed for the previously mentioned wavelength of 400 nanometers, while changes in frequency (expressed in dimensionless units $2\pi c/a$) were introduced by modifying the size of the crystal, and in particular the value of a .

Fig. 3.2 shows the band diagram obtained for TE- and TM-polarized light in a photonic crystal with a square lattice. For neither of these polarizations is a photonic band gap observed. This means that electromagnetic radiation can propagate in such a structure for all frequencies for which the band structure was determined. By cutting out one row of dielectric rods, a waveguide was obtained, and then the propagation of light in it was simulated using the finite element method.

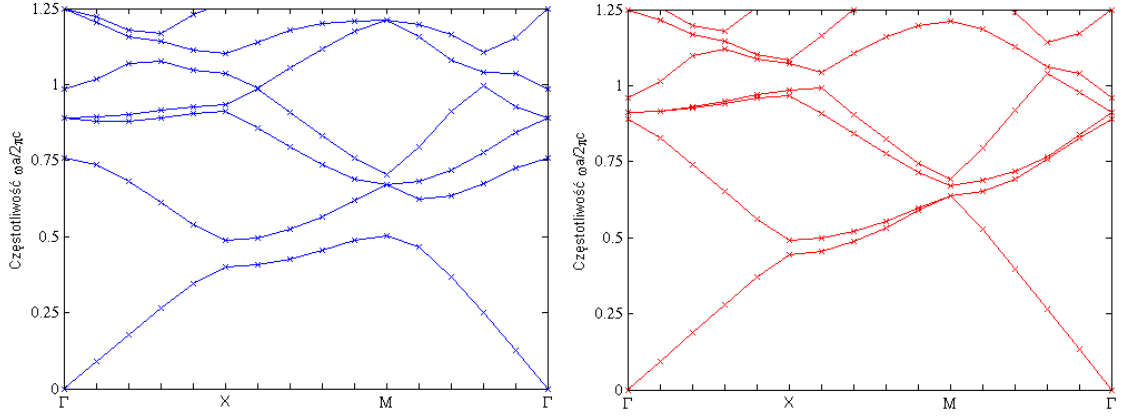


Fig. 3.2. Band structure of a crystal with a square lattice, TE polarization (left plot), TM polarization (right plot).

Figs. 3.3–3.5 show the simulation results obtained for the above case for three different radiation frequencies. In each case, the electromagnetic field was introduced into the system using a hard source located near the considered structure. The simulations were performed for TE polarization. Both the real part of the wave field E_z and its phase are shown. As illustrated schematically in Fig. 2.2, the outer region of the simulation is a PML. All simulations of electromagnetic radiation propagation were performed using the FEM method unless otherwise indicated in the caption.

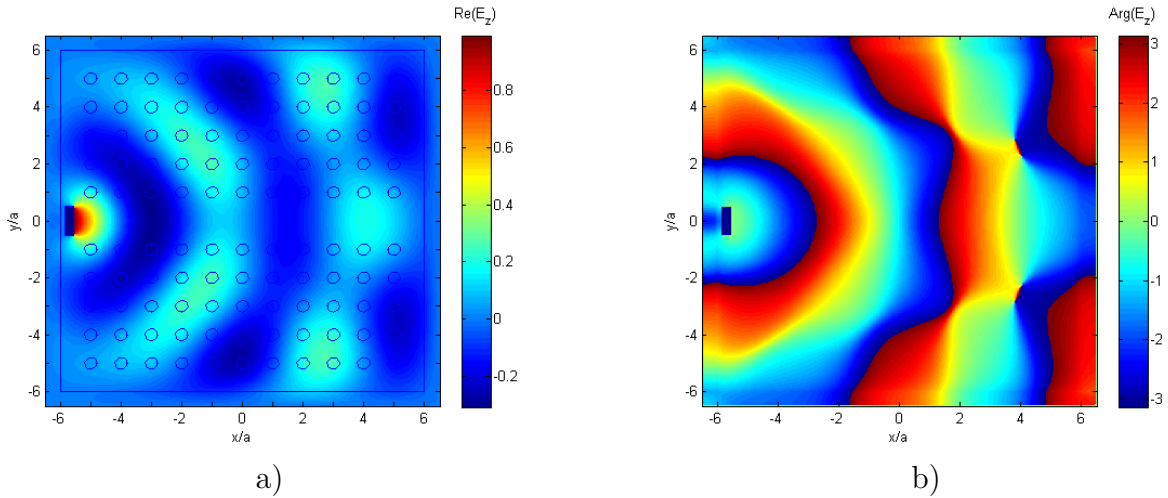


Fig. 3.3. Propagation of an EM wave in a photonic crystal with a square lattice and a linear defect. a) Instantaneous field E_z , b) phase of the field E_z for $\omega = 0.202\pi c/a$.

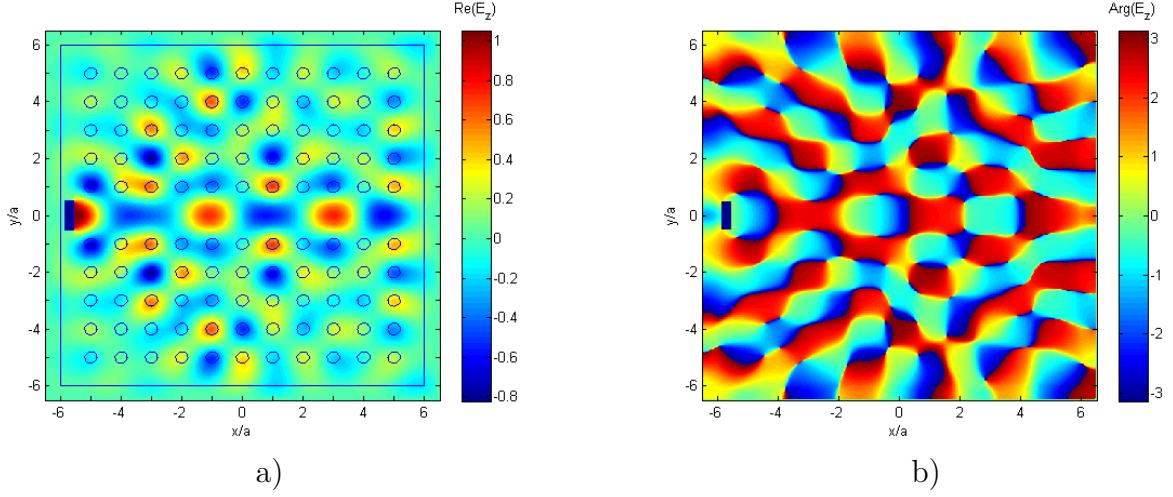


Fig. 3.4. Propagation of an EM wave in a photonic crystal with a square lattice and a linear defect. a) Instantaneous field E_z , b) phase of the field E_z for $\omega = 0.45 2\pi c/a$.

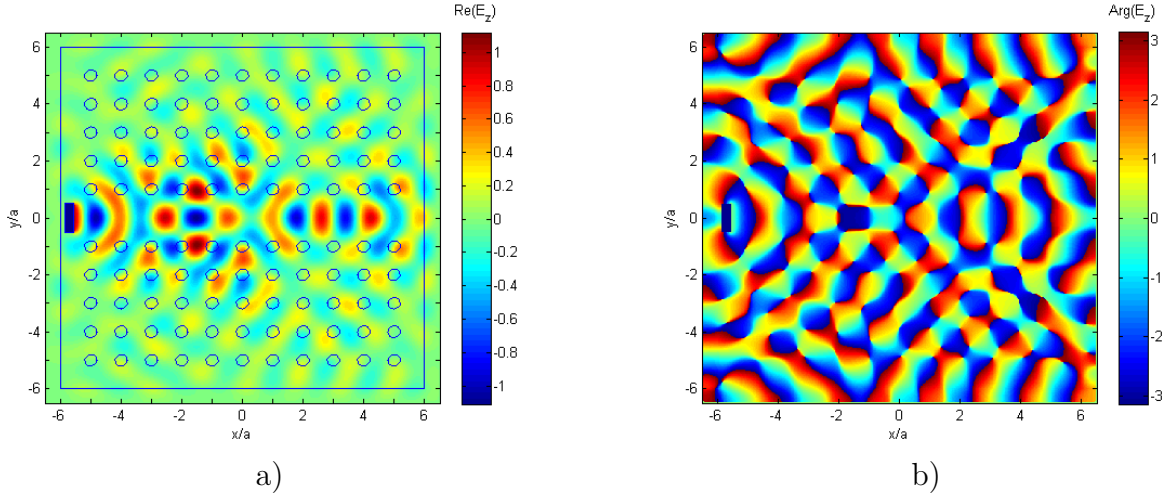


Fig. 3.5. Propagation of an EM wave in a photonic crystal with a square lattice and a linear defect. a) Instantaneous field E_z , b) phase of the field E_z for $\omega = 0.70 2\pi c/a$.

The above simulations confirm that, for the selected frequencies, light propagates in the structure under consideration.

Analogously, the band structure of a photonic crystal with a triangular lattice was determined and is shown in Fig. 3.6. For TE polarization, a band gap is observed in the frequency range $\omega a/2\pi c$ from about 0.48 to about 0.55. It is marked in yellow. In the case of TM polarization, no gap was observed. Therefore, as for the square lattice, the simulations focused only on the first of these polarizations.

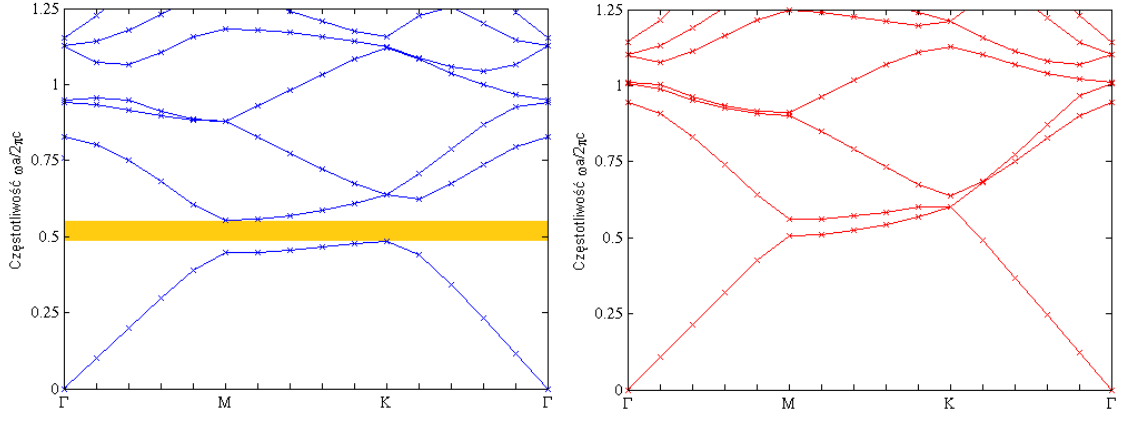


Fig. 3.6. Band structure of a crystal with a triangular lattice, TE polarization (left plot), TM polarization (right plot).

Determining the band structure for a waveguide formed by removing one row of rods indicates the formation of a guided mode (highlighted in green) lying inside the band gap of the crystal, as can be seen in the figure below.

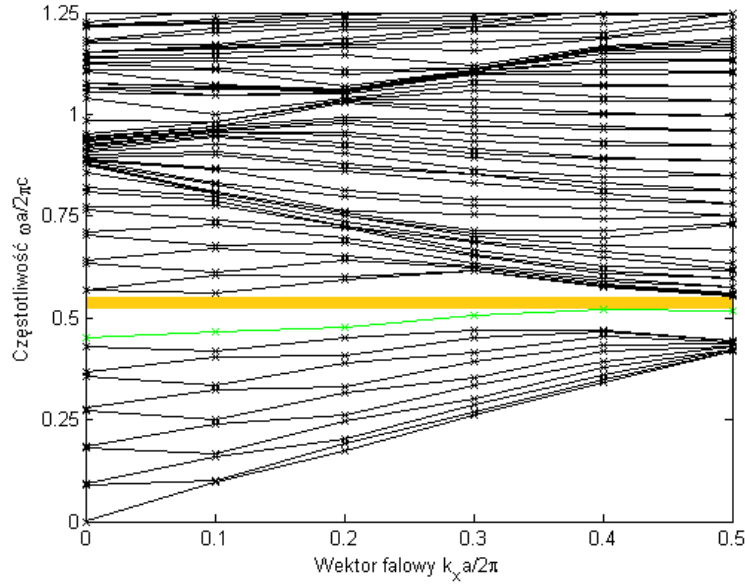


Fig. 3.7. Band structure of a crystal with a triangular lattice and a linear defect.

Figs. 3.8–3.10 show the propagation in such a waveguide of radiation with frequencies $\omega a/2\pi c$ equal to 0.20, 0.45, and 0.70, respectively. They do not lie in the band-gap range, but are intended to illustrate differences between the results obtained for the square and triangular lattices.

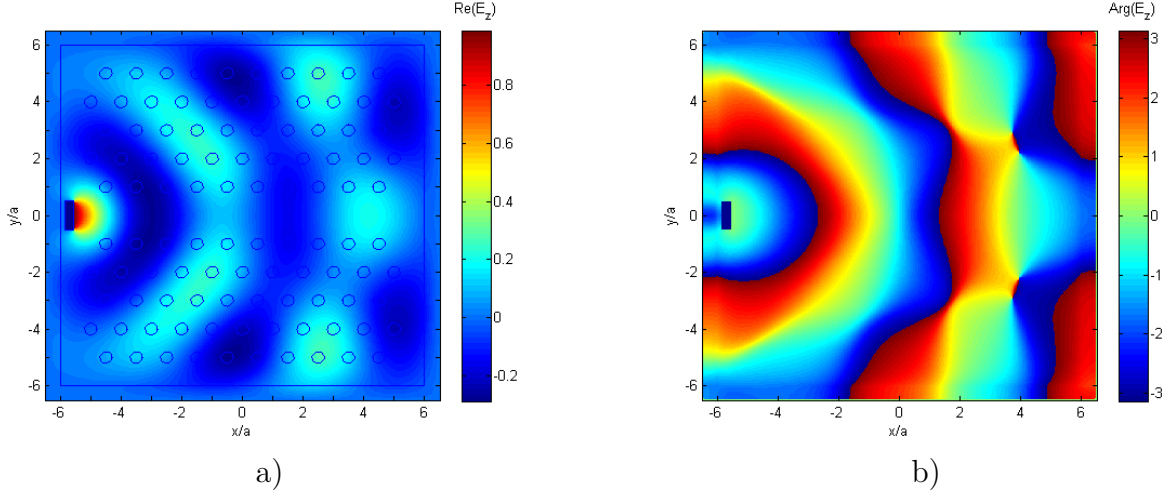


Fig. 3.8. Propagation of an EM wave in a photonic crystal with a triangular lattice and a linear defect. a) Instantaneous field E_z , b) phase of the field E_z for $\omega = 0.20 2\pi c/a$.

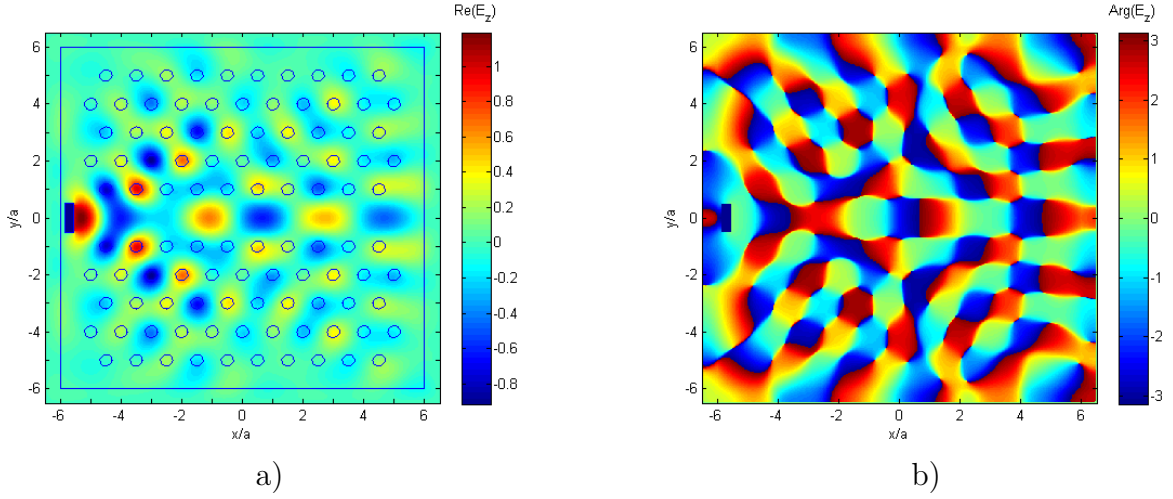


Fig. 3.9. Propagation of an EM wave in a photonic crystal with a triangular lattice and a linear defect. a) Instantaneous field E_z , b) phase of the field E_z for $\omega = 0.45 2\pi c/a$.

Fig. 3.8, obtained for a low frequency, that is, a large wavelength, is consistent with the result obtained for the same input wave when considering the square-lattice system. One can conclude that, in this case, the crystal structure does not affect the way light propagates. This confirms the fact that the lattice constant of a photonic crystal should be comparable with the wavelength with which the crystal is to interact, as mentioned in section 2.5. Only for higher frequencies do differences in propagation begin to be noticeable. In this thesis, however, light propagation was not modeled for very high frequencies, that is, for wavelengths much smaller than the lattice constant of the photonic crystal.

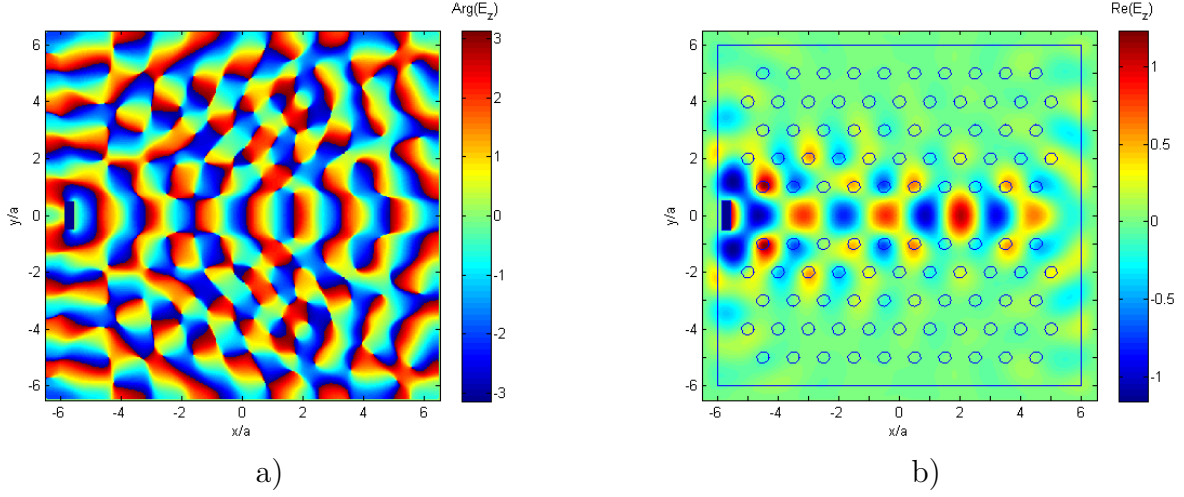


Fig. 3.10. Propagation of an EM wave in a photonic crystal with a triangular lattice and a linear defect. a) Phase of the field E_z , b) instantaneous field E_z for $\omega = 0.702\pi c/a$.

Next, modeling was performed for an electromagnetic wave with a frequency from the guided-mode range. It is easy to see that light propagates mainly along the linear defect, and essentially no propagation in other directions is observed. This is shown below.

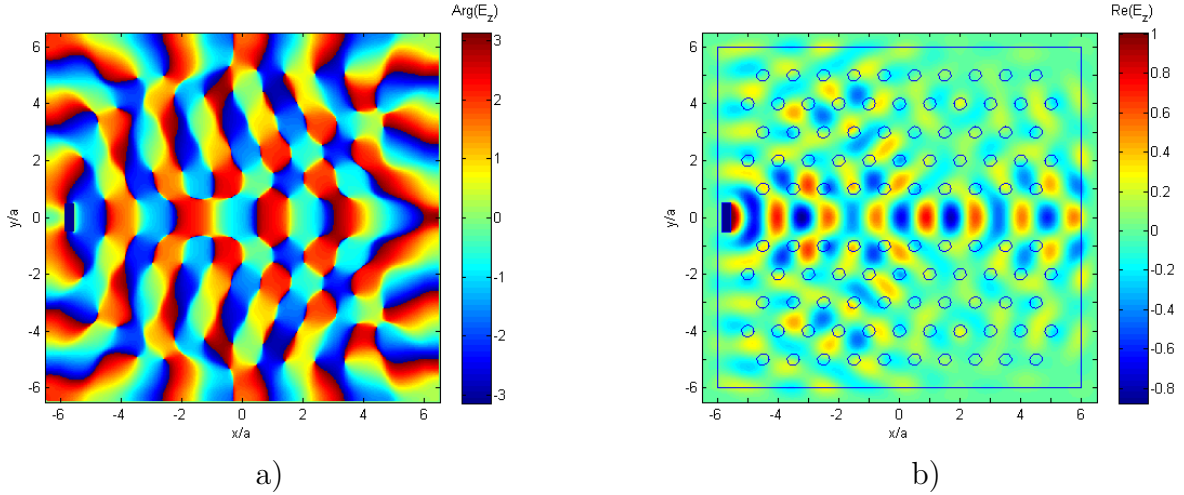


Fig. 3.11. Propagation of an EM wave in a photonic crystal with a triangular lattice and a linear defect. a) Phase of the field E_z , b) instantaneous field E_z for $\omega = 0.502\pi c/a$.

For comparison, the simulation was repeated using the finite difference time domain method. A comparable result was obtained. Small differences may result from the fact that the calculated quantity is the instantaneous field, and also from the fact that a different type of source was used in MEEP, not a hard source, as well as a different input amplitude. There is no doubt, however, that the results obtained by the FEM and FDTD methods agree.

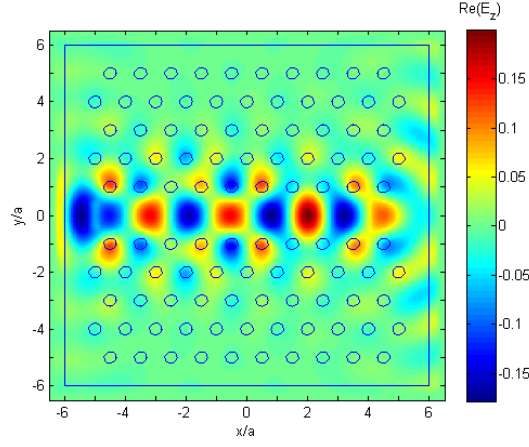


Fig. 3.12. Propagation of an EM wave in a photonic crystal with a triangular lattice and a linear defect. Instantaneous field E_z for $\omega = 0.502\pi c/a$. Simulation performed using the FDTD method.

Finally, for a frequency lying inside the band gap (the value $\omega a/2\pi c = 0.538$ was chosen), a decay of the wave amplitude with increasing distance from the source is observed. The simulation results obtained by the finite element method are therefore consistent with the ranges of the photonic band gap and guided mode determined by the MPB program.

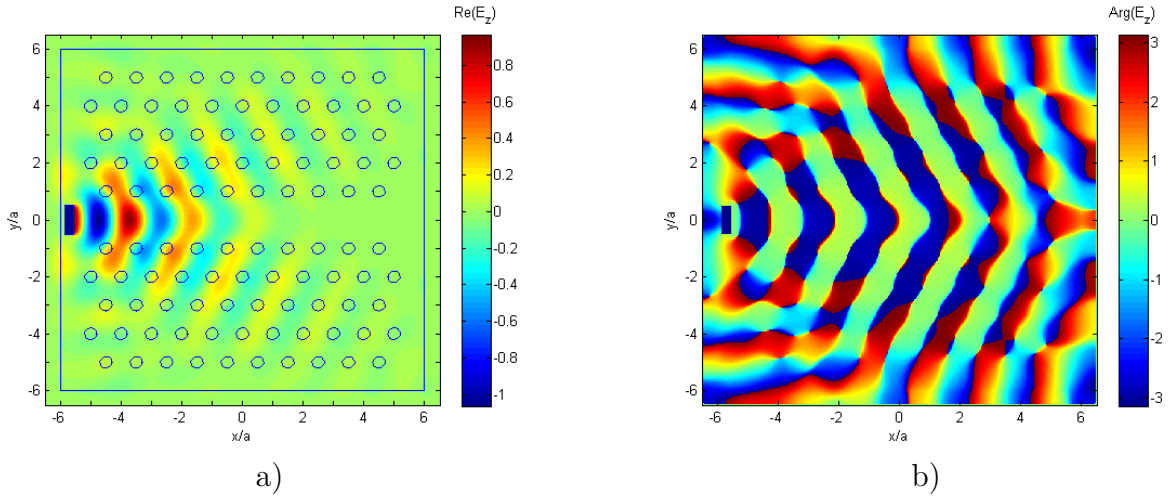


Fig. 3.13. Propagation of an EM wave in a photonic crystal with a triangular lattice and a linear defect. a) Instantaneous field E_z , b) phase of the field E_z for $\omega = 0.5382\pi c/a$.

As in the case of simulations performed for guiding light in a localized defect mode, the finite difference time domain method gives almost identical results to the finite element method for a frequency lying inside the photonic band gap.

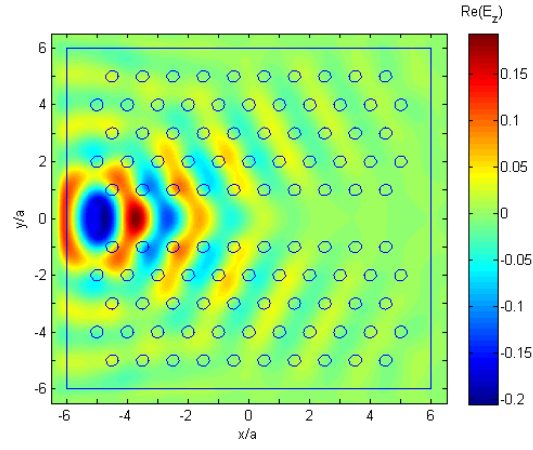


Fig. 3.14. Propagation of an EM wave in a photonic crystal with a triangular lattice and a linear defect. Instantaneous field E_z for $\omega = 0.538\,2\pi c/a$. Simulation performed using the FDTD method.

3.2. Diffractive elements with double periodicity

Diffractive elements with double periodicity are systems built from two subwavelength gratings with different periods. According to what was presented in chapter 2.6, the number of real diffraction orders depends on the direction from which light is incident on such a structure. At the same time, asymmetric transmission of electromagnetic radiation and removal of the zeroth diffraction order can be observed [14].

In the simulations carried out for this chapter, systems made successively of a perfect conductor, chromium, gold, and silver were considered. Their schematic is shown in the figure below.

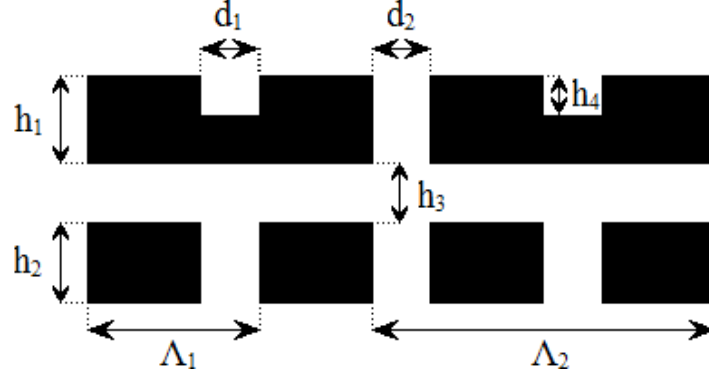


Fig. 3.15. Schematic of a diffraction grating with double periodicity.

The dimensions used were, successively: $2\Lambda_1 = \Lambda_2 = 533$ nm; $d_1 = d_2 = 89$ nm; $h_1 = h_2 = 127$ nm; $h_3 = 89$ nm; $h_4 = 63.5$ nm.

The following simulations were performed using the FEM method for an electromagnetic wave of wavelength 400 nm. TM polarization was used. The electric permittivity values for the metals considered are listed in the table.

Table 3.1 Electric permittivities of selected metals for waves of wavelength 400 nm [15].

	ϵ_1	ϵ_2
Silver	-4.422	0.210
Gold	-1.658	5.735
Chromium	-4.055	11.480

The permittivities ϵ_1 and ϵ_2 contribute to the complex electric permittivity

$$\tilde{\epsilon} = \epsilon_1 + i\epsilon_2. \quad (3.1)$$

Five hard sources, whose centers were located above the slits of the grating, were used to emit spherical waves.

Fig. 3.16 shows the transmission of light through a structure in which the diffraction grating is made of a perfect conductor. For an electromagnetic wave incident from above, two diffraction orders are visible: -1 and 1. For incidence from the other side, only the zeroth diffraction order is observed, which is consistent with the theory presented in section 2.6.

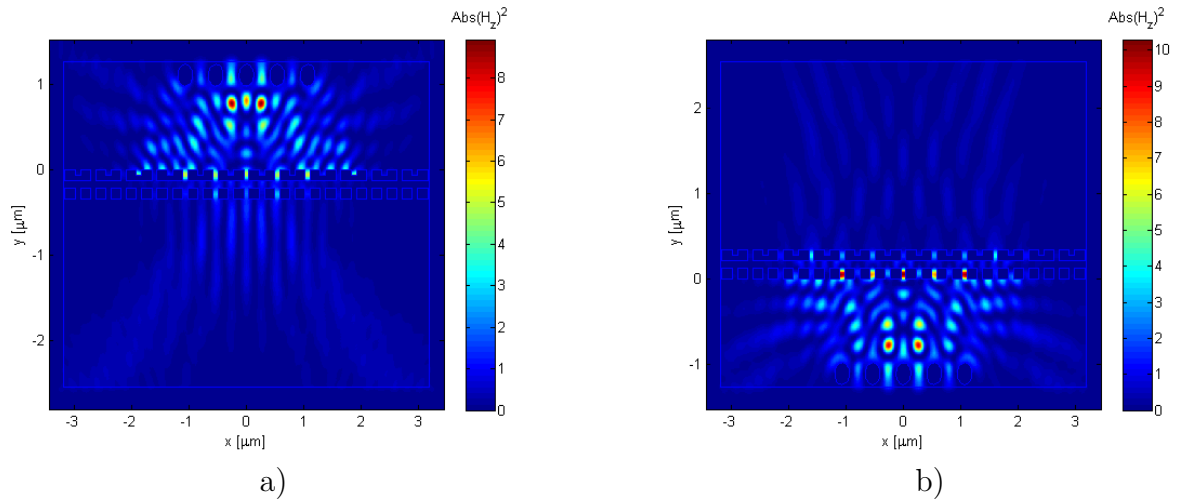


Fig. 3.16. Perfect-conductor grating, a) incidence from above, b) incidence from below.

The use of spherical waves caused a slight deviation of the zeroth diffraction order from the angle 0° with respect to the normal to the direction of periodicity of the structure. This follows from formula 2.29.

Comparable results were obtained for simulations performed for chromium. Increased transmission is observed compared with the perfect conductor, but one can conclude that, of the three metals considered, this one is the most similar to it.

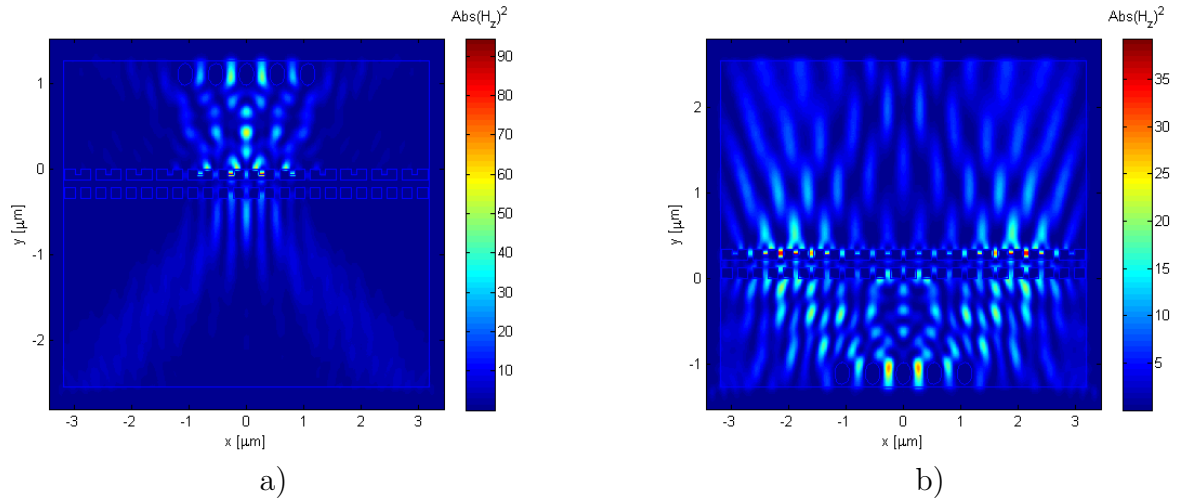


Fig. 3.17. Chromium grating, a) incidence from above, b) incidence from below.

When gold is used, enhanced transmission through the structure is observed. Penetration of light into the elements of the system can be noticed. For the simulation in which the electromagnetic wave was incident from above, three diffraction orders can clearly be seen.

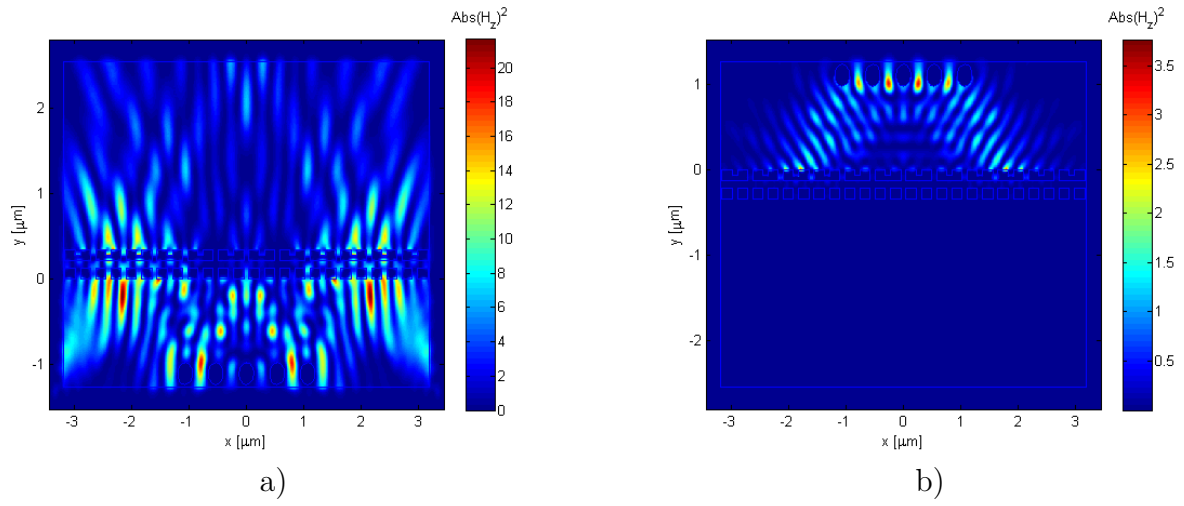


Fig. 3.18. Gold grating, a) incidence from above, b) incidence from below.

Interesting results were obtained for a system made of silver, for which the transmission of electromagnetic radiation turned out to be residual regardless of the direction of incidence.

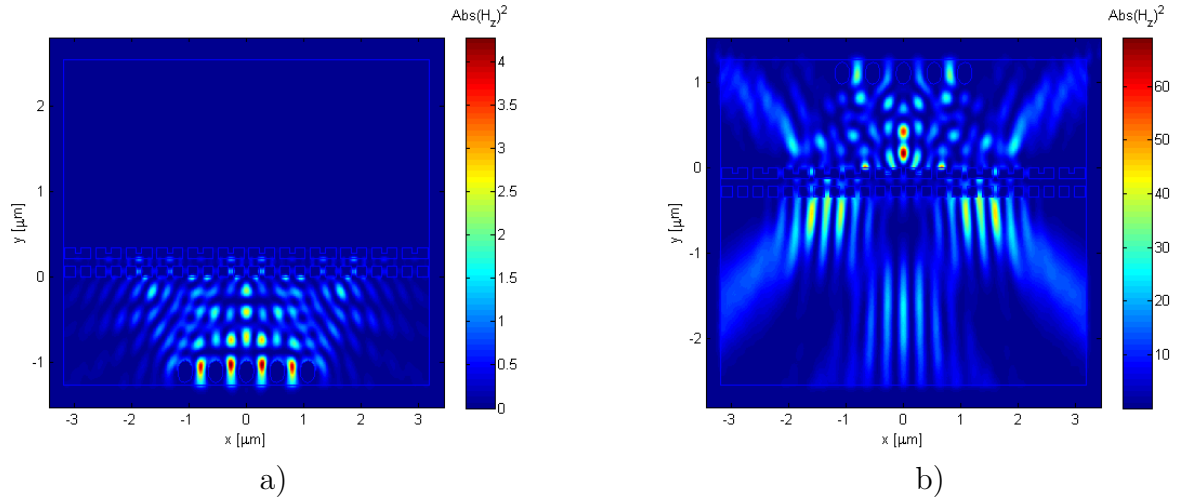


Fig. 3.19. Silver grating, a) incidence from above, b) incidence from below.

Chapter 4. Conclusions

For the purposes of this thesis, a series of simulations concerning the propagation of light in photonic crystals and in diffractive elements with double periodicity were carried out. The work was based on the finite element method.

In the case of photonic crystals, the first step was to determine their band structure, which was possible with the aid of the plane wave method, applicable to optical periodic systems. Two types of lattice were considered: square and triangular. The behavior of light of a specified wavelength was modeled by changing the size of the photonic crystal. It was checked that, in some structures of this type, for certain wave polarizations and in a specified frequency range, a photonic band gap is present. It was also shown that, by introducing defects, efficient guiding of light with a frequency lying inside such a gap becomes possible through the formation of a defect mode. Selected results were then compared with simulations performed using the finite difference time domain method.

For diffractive elements with double periodicity it was shown that the number of observed real diffraction orders generally depends on the direction of illumination of the structure. It was also shown that the material from which the system is made has a significant influence on its operation. For this purpose, modeling was performed for a perfect conductor, chromium, gold, and silver. Many similarities were noted for the perfect conductor and chromium; in the case of gold, enhanced transmission of electromagnetic radiation was observed, while in the case of silver it was almost not observed at all.

It was shown that the finite element method is a useful tool that can be used to design optical systems with characteristic properties. The auxiliary use of tools such as the plane wave method significantly facilitated the simulation of differences in the behavior of light depending on the size of the structure or the material from which it is made. It was demonstrated that combining several techniques for determining the properties of a structure is extremely important in the effective design of new materials. This makes it possible to easily optimize optical elements and also creates an opportunity for the effective design of new, more complex systems.

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